

Physics-Guided PINN Reassessment For Polished Setpoint Forward TE

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

Physics can still help the transmission-error predictor, and the completed Wave 5.2 results do not demonstrate otherwise.

They demonstrate a narrower and more useful result: the specific oscillator, periodic-closure, Bauer-anchor, and quasi-static compliance formulations tested in Phases 2 and 3 did not pass the complete forward curve-first promotion gate. Several constraints were mathematically valid and active during optimization, but they either imposed behavior already present in the data, redistributed error among harmonic orders, introduced an unstable anchor, or produced a seed-dependent predictive benefit.

The correct next strategy is therefore neither to abandon physics nor to add more unqualified loss terms. It is to treat physics as a hierarchy of falsifiable inductive biases:

1. preserve exact structural facts through architecture where possible;
2. use the Polynomial-Fourier law as an inspectable analytical baseline;
3. assign the neural network only the discrepancy that the analytical model cannot explain;
4. supervise the error in curve, derivative, and complex harmonic spaces;
5. balance objectives dynamically and measure gradient conflict;
6. promote a physical ingredient only when it improves the accepted forward baseline on a stable, multi-index, held-out evaluation. This report and its companion roadmap are restricted to:
 - dataset: `polished_dataset` ;
 - input mode: `setpoints` ;
 - surface: `Fw` ;
 - operating coordinates: angle, speed setpoint, torque setpoint, and temperature setpoint;
 - causal deployment target: explicit TwinCAT-compatible inference.No `Bw` , global paired surface, actual-values input, simplified dataset, or paper-faithful MMT experiment is part of this plan.

What A PINN Can Mean In This Project

A physics-informed neural network is not required to contain a complete first-principles model of the reducer. It must, however, contain a precise and testable statement of domain knowledge that changes the learning problem.

Five Levels Of Physics Guidance

Level	Mechanism	Project example	Qualification
Structural representation	The architecture can only produce admissible forms	exact periodic basis in angle	strong inductive bias, not a governing law by itself
Weak physical prior	Penalize implausible signs, bounds, or roughness	positive bounded compliance	physics-guided if the bound has a source and an ablation
Compatibility equation	Enforce a relation among observable quantities	periodic value and slope closure	genuine constraint, but may be redundant
Analytical plus learned residual	Preserve a known model and learn its discrepancy	Bauer Polynomial-Fourier plus neural residual	grey-box physics-informed model
Governing residual	Penalize a differential or algebraic law	oscillator or compliance residual	full PINN when observable and identifiable

This classification prevents two opposite errors

- calling every Fourier feature a physical law;
- rejecting every incomplete physical model because it is not a complete contact solver.

Incomplete Physics Can Be Useful

The most practical formulation is $TE_prediction(\theta, u) = TE_analytical(\theta, u) + residual_neural_network(\theta, u)$, where $u = [speed_setpoint, torque_setpoint, temperature_setpoint]$.

The analytical component supplies a low-complexity explanation of the dominant periodic structure. The network compensates for missing interactions, non-polynomial coefficient surfaces, local ripple, and systematic discrepancy.

This is exactly where a neural network can compensate for an incomplete model. It does not make the analytical model true. It makes the model discrepancy explicit, measurable, and separable.

The main risk is cancellation: an unconstrained residual can learn the negative of the analytical component and turn the system into an opaque black box. Every hybrid experiment must therefore report:

- analytical contribution magnitude;
- residual contribution magnitude;
- correlation between analytical and residual terms;
- residual energy inside and outside the analytical harmonic subspace;
- sensitivity to analytical-anchor corruption;
- performance when the anchor is frozen, partially trainable, or removed.

What The Dataset Actually Supports

Frozen Foundation

The completed Phase 0 audit established

- 969 paired operating conditions;
- 966 Phase 1 eligible conditions after three metadata quarantines;
- 678 / 194 / 97 train, validation, and test conditions before exclusions;
- a stable split signature of
c1aa8718fb9bf88cc2021c121dc4f3b4010fc1d2e45ac90af5f4376aa64f8e16 ;
- a full Cartesian grid of speed, torque, and temperature setpoints;
- one output-equivalent revolution per directional curve;
- causal angle, speed, signed torque, oil temperature, and direction;
- no held-out axis values absent from training;
- no out-of-domain extrapolation condition in the current split.

For this forward-only plan, the split is preserved but only the eligible Fw surface is consumed.

Strong Forward Harmonic Evidence

The Phase 0 forward prevalence ranking begins with orders 1, 39, 3, 40, 78, 240, 81, 159, 2, 156, 162, 80, 120, 237, 42 .

Order 1 has mean truth amplitude close to 0.0172 deg . Orders 39 , 78 , 40 , and 3 form the next strongest forward group in the Phase 2 analysis.

This proves three things

1. the output is strongly periodic in the angular coordinate;
2. a small set of orders contains substantial repeatable information;
3. the amplitude and phase of those orders remain prediction targets whose condition dependence must be learned. It does not prove that every prevalent order maps uniquely to one physical component. Mechanism attribution remains a hypothesis unless the reducer geometry and excitation path support it.

Unsupported Forward-Only Claims

The current scope cannot identify

- directional lost motion from an Fw/Bw difference;
- a backlash gap from forward curves alone;
- a hysteresis state without repeated loops and reset evidence;
- acceleration and inertia effects from nearly steady directional windows;
- contact force, load sharing, clearance, or mesh stiffness;
- input power, internal loss, or efficiency balance;
- wear progression without unit identity and longitudinal history;
- electromechanical sidebands without synchronized motor current;
- MMT equivalent-error groups without condition-varying causal measurements.

These mechanisms remain scientifically relevant. They cannot be inserted as identified forward residuals under the present data contract.

What Was Demonstrated Before This Reassessment

Polynomial-Fourier Analytical Baseline

The Bauer-style law represents TE as a sum of harmonics whose offset, amplitude, and phase vary with operating condition: $TE_PF(\theta, u) = a_0(u) + \sum_k A_k(u) * \sin(k * \theta + \phi_k(u))$.

Each coefficient surface can be a complete quadratic polynomial of the three operating variables. In Phase 1, the selected PF_A_LOCAL_QUADRATIC forward model achieved:

Metric	PF_A_LOCAL_QUADRATIC Fw
Mean raw MAE	0.001807 deg
Mean centered MAE	0.001385 deg
Mean offset error	0.000965 deg

The reduced PF_E_REDUCED_QUADRATIC model was slightly weaker. The recovered ONNX coefficient path achieved 0.003047 deg mean raw MAE on the common forward test surface. The recovered PLC degree-10 polynomial was numerically unsafe across the broader forward torque domain. The direct per-curve Fourier oracle reached approximately 0.00031 deg , proving representational capacity while remaining target-leaking and non-deployable.

The key gap is visible: the deployable quadratic Polynomial-Fourier law leaves substantial distance to the direct Fourier ceiling. This is a rational place for a learned residual.

Harmonic And Kinematic PINN Screen

Phase 2 compared

- **H0** : explicit Fourier data-driven control;
- **H1** : implicit harmonic heads with normalized oscillator residual;
- **H2** : oscillator residual plus periodic value and slope closure;
- **H3** : the same constraints plus a frozen Bauer anchor.

The normalized oscillator residual for each order was $R_k = (1 / k^2) * d2(h_k) / d(\theta)^2 + h_k$.
The validator proved that the residual was correct, differentiable, finite, and active in optimization.
Nevertheless, the forward scalar MAE was:

Candidate	Forward test MAE
H0 Fourier control	0.001646 deg
H2 oscillator plus periodic closure	0.001784 deg
H1 oscillator	0.001951 deg
H3 oscillator, closure, and Bauer	0.002389 deg

On the full held-out forward curves, both accepted references remained stronger:

Candidate	Raw MAE	Amp. error	Phase error
accepted periodic GRU	0.001618	19.237%	12.167 deg
accepted periodic harmonic MLP	0.001694	15.380%	13.592 deg
Phase 2 H0	0.001998	39.328%	21.029 deg
Phase 2 H1	0.002232	43.016%	19.899 deg
Phase 2 H2	0.002102	49.491%	20.335 deg
Phase 2 H3	0.002651	69.329%	27.110 deg

The periodic-slope term reduced its own residual by several orders of magnitude. The original curves already had approximately 0.0009 deg closure mismatch, so optimizing closure added little useful information.
This is a textbook case of a valid constraint being redundant rather than wrong.

Quasi-Static Compliance PINN Screen

Phase 3 tested learned-mean controls, soft compliance derivatives, temperature-conditioned stiffness, nonlinear compliance, hard elastic equations, and shared-stiffness compatibility.

The best forward candidate, `C1`, used $d(\text{mean_TE}) / d(\text{tau_signed}) = 1 / k_{\text{forward}}$.

with positive bounded stiffness. On the forward full-curve screen:

Candidate	Raw MAE	Offset	Centered MAE	Amp. error	Phase
accepted periodic GRU	0.001618	0.000690	0.001382	19.237%	12.167 deg
accepted periodic MLP	0.001694	0.000825	0.001390	15.380%	13.592 deg
C1	0.001843	0.000932	0.001481	34.128%	21.434 deg
C0 control	0.001953	0.000989	0.001608	40.930%	22.373 deg

`C1` improved every listed metric except peak-to-peak error relative to its matched `C0` control. Its fitted stiffness was also stable:

- mean: 28,164.36 Nm/deg ;
- population coefficient of variation: 2.48% .

However, only two of three initializations passed the predeclared joint gate. The physical parameter was stable while the predictive gain was not. This is important positive information. Compliance is not disproved. The tested joint optimization is not yet reliable enough to become a default ingredient.

Why Valid Physics Did Not Automatically Improve Prediction

Redundant Information

If the architecture already uses exact Fourier heads, an oscillator residual may merely restate the definition of those heads. If the measured curve is already nearly periodic, a stronger closure loss spends gradient budget on an almost-solved objective.

The next experiments must quantify the incremental information in each constraint before training.

Loss-Scale And Gradient Conflict

PINNs are multi-objective optimizers. A small reported residual does not imply that the residual helped the data objective. Different terms can have:

- incompatible gradient directions;
- orders-of-magnitude gradient differences;
- different convergence rates;
- different noise sensitivities;
- incompatible optima under model misspecification.

The new roadmap therefore requires per-loss gradient norms, cosine similarities, and weight trajectories. Fixed physics weights will be controls, not the only strategy.

Model Misspecification

The Bauer law is semi-analytical and incomplete. Compliance is only one component of mean TE. Harmonic orders may be correct while their coefficient surfaces are too rigid.

A hard or strongly weighted imperfect model can bias the network away from the measured surface. The solution is not to hide the mismatch; it is to model it explicitly through a bounded discrepancy branch.

Identifiability

Low prediction error does not prove that a learned stiffness, phase law, or latent state is physically correct. Multiple decompositions can reconstruct the same curve.

The roadmap separates

- predictive promotion;
- parameter stability;
- mechanism identification;
- deployment usefulness.

A model may be accepted for prediction without claiming that every latent quantity is a uniquely identified physical parameter.

Spectral Competition

The Phase 2 results showed improvement in some harmonic orders and regression in others. Aggregate pointwise loss can hide this redistribution.

The new loss design must distinguish

- low-order shape;
- reducer-related middle orders;
- fragile high-order ripple;
- offset;
- complex amplitude and phase;
- residual energy outside the selected basis.

Candidate Physics-Guided Formulations

A. Polynomial-Fourier Plus Bounded Residual

The proposed prediction law is $\text{TE}_{\text{hat}} = \text{TE}_{\text{PF-A}} + \alpha(u) * r_{\text{theta}}(\text{theta}, u)$.

where $\alpha(u)$ is bounded and the residual network is capacity-limited.

Required ablations

- pure PF-A ;
- identical residual network without PF-A ;
- frozen PF-A plus residual;
- partially trainable coefficient correction;
- corrupted PF-A anchor plus residual;
- residual with and without analytical-subspace exclusion.

This is the highest-priority family because every inference quantity is available and the analytical baseline is already reproduced.

B. Coefficient-Residual Model

Instead of correcting the curve pointwise, predict corrections to offset and complex harmonic coefficients:

$$c_k(u) = c_{k_PF}(u) + \delta c_k(u), \text{ with } c_k = a_k - i * b_k.$$

Using sine and cosine or complex coefficients avoids phase-wrap discontinuity. Amplitude and phase can be derived for reporting.

Advantages

- exact periodic reconstruction;
- lower-dimensional learned output;
- inspectable condition dependence;
- direct harmonic-band regularization;
- straightforward PLC realization.

Risks:

- retained orders may omit local structure;
- coefficient errors can interact nonlinearly in amplitude and phase;
- a direct coefficient target must be computed from training curves only.

C. Mean Plus Centered-Shape Decomposition

The prediction is $TE_{\hat{}}(\theta, u) = \text{mean_head}(u) + \text{centered_shape_head}(\theta, u)$, with $\int \text{centered_shape_head} d\theta = 0$.

The zero-mean property should be imposed by construction, not by a large loss. This separates offset learning from angular shape learning.

Candidate auxiliary outputs

- mean TE;
- peak-to-peak TE;
- selected complex harmonic coefficients;
- derivative or curvature summaries;
- bounded compliance slope as a diagnostic head.

D. Weak Compliance Guidance

Compliance should return as a weak, staged prior rather than an immediate hard equation. The proposed loss is $L_{\text{compliance}} = \text{penalty_negative}(d \text{ mean_TE} / d \tau) + \text{penalty_outside_source_bound}(k_{\text{effective}})$.

The exact derivative target should not be forced uniformly across all conditions unless supported by the data. The candidate can use:

- sign-only monotonicity;
- broad stiffness interval;
- temperature-stratified interval;
- confidence-weighted penalty;
- warm-started activation after data convergence.

E. Spectral And Sobolev Objectives

For each held-out or training curve

- $L_{\text{complex}_k} = \text{abs}(c_k_{\text{prediction}} - c_k_{\text{truth}})^2$;
- $L_{\text{slope}} = \text{mean}(\text{abs}(dTE_{\text{prediction}}/d\theta - dTE_{\text{truth}}/d\theta))$;
- $L_{\text{band}} = \sum_k w_k * L_{\text{complex}_k}$.

This is supervised shape guidance, not a governing physical law. It is nevertheless valuable because it targets the exact failure mode observed in Phase 2. The physically informed element comes from the selection and grouping of orders, the analytical basis, and any source-backed coefficient constraints.

F. Temporal Residual Model

The accepted forward GRU remains the strongest time-windowed reference. A temporal hybrid can use

$$TE_{\text{hat}}_t = TE_{\text{PF}}(\theta_t, u_t) + GRU_{\text{residual}}(\text{history}_t) .$$

This is appropriate only if the history inputs remain causal and no target-derived state is used at inference. It should be compared against:

- the accepted GRU;
- a parameter-matched residual GRU without analytical anchor;
- a non-windowed residual MLP.

G. Conditional Periodic Implicit Representation

Fourier-feature MLP and SIREN-style variants can represent high-frequency coordinate functions and their derivatives. They are candidates for the centered-shape or residual branch, not physical laws.

They should be tested only after the explicit harmonic models because:

- explicit orders are more inspectable;
- periodic activations can create unwanted frequencies;
- initialization and bandwidth are critical;
- deployment cost may exceed a compact coefficient model.

H. Weak-Form Harmonic Residual

A variational residual integrates the equation against test functions instead of enforcing high-order derivatives pointwise: $R_j = \text{integral}(R(\theta, u) * v_j(\theta) d\theta)$.

For the TE problem, test functions can be localized Fourier or polynomial windows. This may reduce derivative noise and reveal where a pointwise oscillator constraint is redundant.

It should be evaluated as an ablation against direct complex-coefficient loss. If both encode the same information, the simpler coefficient loss wins.

I. Sparse Residual Discovery

Sparse identification can screen a controlled dictionary of condition and harmonic interactions. The initial library contains `1` , `tau` , `speed` , `temperature` , their squares and pairwise products, `sin(k*theta)` , `cos(k*theta)` , and condition-harmonic interaction terms.

The goal is not to declare a newly discovered physical law from one dataset. It is to find parsimonious candidate terms for a later interpretable model.

Selection must occur on training conditions only and be validated on held-out conditions.

Deep Technique Discovery

The following external techniques were reviewed for project relevance.

Technique	Main idea	Forward TE relevance	Priority
Gradient-statistics annealing	balance composite loss gradients	directly addresses Phase 2 and 3 objective imbalance	high
ReLoBRaLo or related adaptive weighting	weight objectives by relative learning progress	useful for curve, spectral, and weak-physics losses	high
Gradient conflict diagnostics or PCGrad	detect or project conflicting task gradients	useful before accepting any multi-head benefit	high
Curriculum regularization	activate difficult physics progressively	avoids early domination by imperfect residuals	high
Model-discrepancy PINNs	learn an explicit correction to misspecified physics	direct match to Bauer plus residual	high
Fourier features	increase learnable angular bandwidth	useful residual or centered-shape ablation	medium-high
SIREN	periodic activations with accurate derivatives	useful coordinate-network ablation	medium
Sobolev training	supervise derivatives as well as values	matches slope and ripple fidelity objectives	high
VPINN or weak form	enforce integrated residuals with lower derivative order	useful oscillator and local-band ablation	medium
Failure-informed sampling	focus collocation on high-residual regions	useful only if angular or condition errors are localized	medium
Augmented Lagrangian constraints	separate constraint satisfaction from fixed penalties	useful for exact closure or bounds if soft loss fails	medium
Universal differential equations	replace missing terms in a known differential model	strong future pattern; current steady F_w law is limited	medium-low
Sparse equation discovery	select parsimonious interaction terms	useful offline formulation discovery	medium
Bayesian or ensemble discrepancy	quantify trust in physics and residual branches	useful near domain boundaries and for deployment	medium
Neural operators	learn maps between function spaces	possible only with a richer curve-to-curve input contract	low now
Formal residual certification	bound worst-case continuous residual	valuable after a candidate exists, not an initial model	later

Research Interpretation

The literature reinforces five lessons.

1. Composite PINN losses commonly suffer from gradient imbalance.
2. Harder physics does not guarantee better optimization.
3. High-frequency behavior requires architectural and loss-level attention.
4. Imperfect physics should be paired with an explicit discrepancy model.
5. Residual satisfaction and predictive accuracy must be validated separately.

These lessons align closely with the local Wave 5.2 evidence.

Techniques Deliberately Excluded From The Active Forward Plan

Technique or mechanism	Reason for exclusion
Directional backlash gap	requires bidirectional or loop evidence
Bouc-Wen or rolling-friction state	no repeated forward loops or reset contract
Dynamic inertia residual	current forward windows are nearly steady and inertia is unavailable
Contact and mesh-force PINN	contact state, stiffness, clearance, and force are unavailable
Energy-balance PINN	input power and internal losses are unavailable
MMT paper-faithful residual	condition-varying equivalent-error inputs are unavailable
Wear or degradation state	no longitudinal reducer identity or load-cycle history
Electromechanical residual	no synchronized motor current or sideband evidence
Forward neural operator	no function-valued causal input justifying the additional complexity

Exclusion means “not identifiable in this data contract,” not “physically invalid.”

Revised Experimental Logic

Separate Representation From Physics

Every candidate must declare which improvement is being tested:

- representation capacity;
- analytical prior;
- weak physical constraint;
- optimization strategy;
- temporal information;
- uncertainty model.

No experiment may change all of them simultaneously.

Use A Ladder Of Controls

For every hybrid

1. accepted periodic GRU;
2. accepted periodic harmonic MLP;
3. pure analytical PF-A ;
4. parameter-matched black-box;
5. analytical plus residual without physics loss;
6. analytical plus residual with one physics ingredient.

This ladder tells us whether a gain comes from physics, extra parameters, or a better representation.

Require Stable Evidence

Every promotable training configuration requires

- at least three fixed seeds;
- identical condition split and angular evaluation;
- no target-derived inference input;
- complete checkpoint and loss-weight provenance;
- no unreported failed run;
- parameter and gradient diagnostics;
- full-resolution forward curve replay.

Measure More Than MAE

Required forward metrics

- raw MAE and RMSE;
- absolute mean-offset error;
- mean-centered MAE;
- peak-to-peak relative error;
- derivative MAE and correlation;
- complex coefficient error by selected order;
- amplitude and phase error;
- P95 per-curve error;
- worst operating-cell degradation;
- residual-to-anchor energy ratio;
- seed dispersion;
- inference cost and numerical range.

Add Boundary Tests

The current held-out split is interpolation over supported axis values. Additional evaluation should create training-only boundary holdouts such as:

- maximum and minimum torque bands;
- maximum and minimum speed bands;
- temperature-level holdout;
- corner-cell holdouts;
- harmonic-amplitude extremes.

These are experimental splits, not replacements for the canonical split.

Answer To The Central Question

Yes: the neural network can compensate for missing parts of an incomplete physical model.

It will help only when the physical component contributes information that is useful, correctly scaled, observable, and not already encoded by the network. The network must then have enough freedom to model the discrepancy without being allowed to erase the analytical structure invisibly.

The completed tests were valuable because they exposed exactly where naive physics insertion fails:

- mathematically true but redundant constraints;
- correct local effects that do not improve the complete curve;
- unstable optimization across seeds;
- physical parameters that are stable without sufficient predictive gain;
- mechanisms that are real but unobservable in the available data.

The next full-PINN effort should therefore begin with the strongest observable physics: angular periodic structure, condition-dependent harmonic coefficients, an explicit Polynomial-Fourier anchor, and carefully bounded forward compliance. It should combine them with discrepancy learning, frequency-resolved supervision, and modern multi-objective optimization.

Program Decision

The previous sixteen-phase Wave 5.2 evidence closeout remains valid. This report does not retroactively promote a rejected residual.

It creates the new forward-only `Wave 5.2R: Physics-Guided Forward Reassessment` branch for `polished_dataset / setpoints / Fw`.

Wave 5.2R is currently a documented roadmap, not an authorized training campaign. Each bounded implementation or campaign stage requires its own technical document, campaign plan where applicable, explicit approval, and curve-first closeout.

Local Evidence

- `doc/reference_summaries/10_Polynomial_Fourier_TE_Model_Project_Summary.md`
- `doc/reference_summaries/11_Hysteresis_Backlash_And_Harmonic_TE_Reference_Synthesis.md`
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